Urban Air Pollution and Time Losses: Evidence from Cyclists in London — Revisited

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Abstract

We replicate and extend Klingen and van Ommeren (2020), who found that a 10 ppb increase in ozone reduces cycling speed by 0.3–0.4% using 42 million London bike-share trips (2013–2017). Extending the sample to 112 million trips through 2025, we find a negative ozone–speed relationship in all four identification strategies. The within-day estimate is stable across periods, at about 0.05 km/h per 10 ppb. The within-cyclist estimate, based on pseudo-cyclists defined within a calendar year-month, is about half the original estimate and smaller after 2017, but remains statistically significant.

Keywords: air pollution, ozone, physical effort, cycling, productivity, replication
JEL Codes: I18, J24, Q53, R41

1 Introduction

Klingen and van Ommeren (2020) provided novel evidence that ambient ozone reduces cycling speed, using 42 million Santander Cycles trips during 2013–2017. Their preferred within-day estimate implied that a 10 ppb increase in ozone reduces speed by about 0.4%, with effects emerging at concentrations as low as 20 ppb—far below regulatory standards.

We revisit their analysis with two objectives. First, we replicate the original results on the 2013–2017 sample, implementing the same four identification strategies (within-day, between-day, spatio-temporal, cyclist panel) without traffic controls, which the original study showed to be inconsequential. Second, we extend the analysis to 2013–2025, with 112 million trips. We also estimate models on the post-2017 subsample alone (2018–2025) to test whether the relationship holds in a period that includes COVID-19, the Ultra Low Emission Zone expansion, and substantial cycling infrastructure changes.

Our replication confirms the core findings. The within-day ozone coefficient is similar in the replication, extended and post-2017 samples. The between-day and spatio-temporal strategies yield negative coefficients, although the spatio-temporal estimate is imprecise for 2013–2017. The cyclist-panel estimate is smaller than in the original study, which we attribute to a stricter definition of pseudo-cyclists.[1] Robustness checks—non-linearity analysis, lag/lead tests, and median speed regressions—support the original conclusions.

2 Methods and Data

We implement the four identification strategies of Klingen and van Ommeren (2020); see their paper for details. All models are estimated on weekdays only.

Within-day. Hourly city-level speed is regressed on ozone (per 10 ppb), flexible weather controls (temperature dummies, pressure, humidity, rain, radiation dummies, wind, 8-hour temperature and rain lags), and co-pollutant controls (SO_2, NO_x and $PM_{2.5}$ dummies), with day and day-of-week×hour fixed effects. WLS with trip counts as weights; SEs clustered by date.

Between-day. Daily aggregates with year-month and day-of-week fixed effects.

Spatio-temporal. Zone-day aggregates (5 monitoring zones, 2 km radius; a trip is assigned to a zone if both its origin and destination station are within 2 km of the monitor), with

[1] The original study pooled the same calendar month over years, so that a pseudo-cyclist could combine trips made several years apart.

day, zone, and zone-month (or zone-week) fixed effects and NO_x dummies.

Cyclist panel. Pseudo-cyclists are identified from repeat trips on the same route (origin and destination station), in the same clock hour and the same calendar year-month, with at most one trip per day, at least 4 trips, and departure-time regularity below the 5th percentile of the random uniform distribution. Unlike the original study, which pooled the same calendar month over all years, a pseudo-cyclist is confined to one year-month. A check on New York's Citi Bike, where trips of members in 2017–2019 report the rider's birth year and gender, suggests that this makes pseudo-cyclists considerably purer: within a pseudo-cyclist, 88% of trip pairs share birth year and gender, against 80% when the calendar month is pooled over years (2% for two random members). Electric bikes, introduced in September 2022 and about 2 km/h faster than classic bikes for the same pseudo-cyclist, are excluded. Cyclist fixed effects absorb route and time-of-day preferences.

Data. Trip data are from Transport for London's Santander Cycles. TfL reports clock times in GMT until mid-September 2013 and in UK local time thereafter; we convert all times to GMT to match the air quality and weather data. Air quality data (O_3, NO_2, NO, $PM_{2.5}$, SO_2) are from the London Air Quality Network (5 stations). Weather data are from the Open-Meteo archive. The replication sample covers 2013–2017 (≈41M trips), the extended sample 2013–2025 (≈112M trips), and the post-2017 sample 2018–2025 (≈70M trips).

Table 1 reports descriptive statistics. Mean ozone is higher in the post-2017 period (25.2 vs. 18.7 ppb) while NO_x has declined substantially (21.8 vs. 44.6 ppb), consistent with ULEZ effects. Mean cycling speed is slightly higher in recent years (12.5 vs. 12.0 km/h).

Table 1: Descriptive statistics (hourly, trip-weighted).

Panel A: Replication (2013–2017).

Statistic	Mean	St Dev.	Min	Max
Speed (km/h)	11.98	1.67	5.85	17.75
Distance (km)	2.82	0.29	1.40	5.36
Duration (sec)	1110.40	325.03	443.33	10658.68
Number of trips	1821.43	1082.04	4.00	6893.00
Ozone (O_3, ppb)	18.69	11.65	0.04	78.28
Nitrogen oxides (NO_x, ppb)	44.60	33.83	1.38	509.24
Particulate matter ($PM_{2.5}$, $\mu g/m^3$)	13.23	10.17	-2.60	114.87
Sulphur dioxide (SO_2, ppb)	1.05	0.88	-0.29	92.72

Panel B: Post-2017 (2018–2025).

Statistic	Mean	St Dev.	Min	Max
Speed (km/h)	12.47	1.65	1.24	22.72
Distance (km)	3.06	0.33	0.96	11.56
Duration (sec)	1118.89	299.41	247.42	24468.88
Number of trips	1865.29	1057.95	1.00	6642.00
Ozone (O_3, ppb)	25.22	12.75	0.07	99.77
Nitrogen oxides (NO_x, ppb)	21.85	18.87	0.28	270.19
Particulate matter ($PM_{2.5}$, $\mu g/m^3$)	8.88	7.33	-2.65	151.62
Sulphur dioxide (SO_2, ppb)	0.59	0.63	-0.69	52.24

Means and standard deviations weighted by number of trips per hour.

3 Results

3.1 Within-day estimation

Table 2 presents the within-day results. In the replication sample, a 10 ppb increase in ozone reduces speed by 0.047 km/h (column 1), close to the original estimate of -0.053. The effect is similar in the full extended sample (-0.057, column 2) and the post-2017 subsample (-0.050, column 3), all significant at the 1% level.

The sorting tests show small effects of ozone on trip distance (columns 4–6) and log trips (columns 7–9). In the extended and post-2017 samples, a 10 ppb increase reduces mean distance by about 0.009 km (0.3%) and raises the number of trips by 0.2–0.3%. These effects are too small to suggest that the speed effect is driven by compositional changes.

Table 2: Within-day estimation.

Dependent Variables:	mean_speed			mean_dist			
	Speed (R)	Speed (E)	Speed (P)	Dist (R)	Dist (E)	Dist (P)	Trips (R)
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
O3_10	-0.0472***	-0.0568***	-0.0497***	0.0007	-0.0089***	-0.0082**	0.0001
	(0.0114)	(0.0105)	(0.0127)	(0.0026)	(0.0025)	(0.0032)	(0.0009)
Fixed-effects							
date	Yes	Yes	Yes	Yes	Yes	Yes	Yes
dow_hour_daylight	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	31,092	78,974	47,882	31,092	78,974	47,882	31,092
R^2	0.93296	0.92172	0.92573	0.92668	0.91425	0.91177	0.99850
Within R^2	0.20249	0.11067	0.12719	0.40627	0.31654	0.37065	0.98830

Clustered (date) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

WLS with trip counts as weights. SEs clustered by date. (R) = 2013–2017, (E) = 2013–2025, (P) = 2018–2025.

3.2 Between-day estimation

Table 3 reports between-day results. The replication coefficient (-0.246, column 1) is larger than the within-day estimate, consistent with the original study and with the fact that between-day variation in ozone may capture additional cumulative exposure effects. The post-2017 estimate (-0.076, column 3) is smaller but remains significant at the 5% level. The distance and trip-count coefficients are small and mostly insignificant.

Table 3: Between-day estimation.

Dependent Variables:	mean_speed			mean_dist			
	Speed (R)	Speed (E)	Speed (P)	Dist (R)	Dist (E)	Dist (P)	Trips (R)
Model:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
O3_10	-0.2463***	-0.1113***	-0.0758**	0.0114*	0.0037	0.0003	-0.0032*
	(0.0539)	(0.0271)	(0.0323)	(0.0060)	(0.0033)	(0.0040)	(0.0018)
Fixed-effects							
yearmonth	Yes	Yes	Yes	Yes	Yes	Yes	Yes
dow	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics							
Observations	1,302	3,349	2,047	1,302	3,349	2,047	1,302
R^2	0.53956	0.68250	0.73633	0.80746	0.90030	0.87633	0.99528
Within R^2	0.29516	0.21012	0.23082	0.56991	0.48002	0.45433	0.98756

Clustered (date) standard-errors in parentheses

*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

WLS with daily trip counts as weights. SEs clustered by date. Fixed effects: year-month and day-of-week.

3.3 Spatio-temporal estimation

Table 4 reports the spatio-temporal results. In the replication sample, the zone-month specification yields a near-zero coefficient (column 1), while the zone-week specification gives -0.049 (column 2), both imprecisely estimated—consistent with the wide confidence intervals in the original study. In the extended sample, the zone-week estimate is -0.075 and significant at the 1% level (column 4). In the post-2017 sample, both specifications yield negative coefficients (-0.047 and -0.089), with the zone-week estimate significant at the 5% level (column 6).

Table 4: Spatio-temporal estimation (dependent variable: speed).

Dependent Variable:	mean_speed					
	Speed-M(R)	Speed-W(R)	Speed-M(E)	Speed-W(E)	Speed-M(P)	Speed-W
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
O3_10	0.0052	-0.0491	-0.0329	-0.0747***	-0.0474	-0.0889
	(0.0414)	(0.0458)	(0.0261)	(0.0282)	(0.0341)	(0.0358
Fixed-effects						
date	Yes	Yes	Yes	Yes	Yes	Yes
zone	Yes	Yes	Yes	Yes	Yes	Yes
zone_month	Yes		Yes		Yes	
zone_week		Yes		Yes		Yes
Fit statistics						
Observations	5,891	5,873	14,352	14,313	8,461	8,440
R^2	0.95278	0.96972	0.95864	0.97365	0.96100	0.9752
Within R^2	0.02107	0.01780	0.00761	0.00894	0.00825	0.0124

Clustered (date) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

WLS with zone-day trip counts as weights. SEs clustered by date. M = zone-month FE, W = zone-week FE.

3.4 Cyclist panel estimation

Table 5 presents the cyclist panel results across all three periods. The panel contains 836,027 pseudo-cyclists and 6.0 million trips. The replication coefficients (-0.014 and -0.016) are about half the original study's range of -0.026 to -0.031 (their Table 4), which plausibly reflects our stricter definition of pseudo-cyclists, confined to one year-month. The extension and post-2017 coefficients are smaller still (-0.009 and -0.007 for all distances), but remain statistically significant, so the within-cyclist ozone effect persists.

Table 5: Cyclist panel estimation.

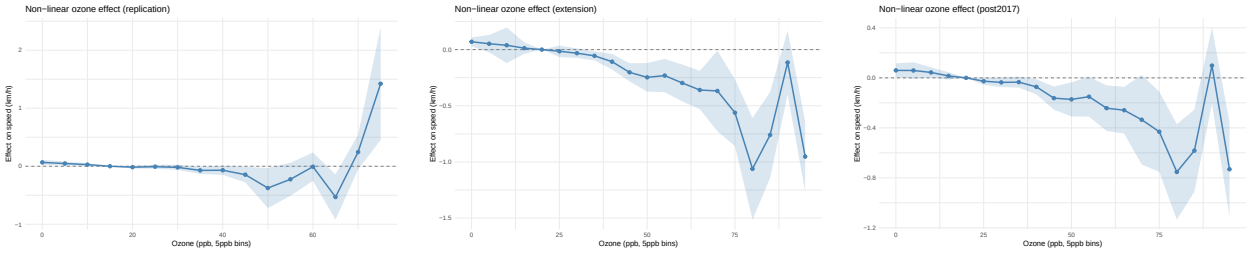
Dependent Variable:	speed_kmh					
Model:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
O3_10	-0.0138***	-0.0155***	-0.0090***	-0.0101***	-0.0065**	-0.0075**
	(0.0031)	(0.0026)	(0.0021)	(0.0019)	(0.0028)	(0.0028)
Fixed-effects						
cyclist_set	Yes	Yes	Yes	Yes	Yes	Yes
dow	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
Observations	2,400,166	1,791,403	5,543,683	4,174,014	3,143,517	2,382,611
R^2	0.77774	0.80148	0.79147	0.81203	0.79973	0.81791
Within R^2	0.00281	0.00357	0.00263	0.00333	0.00314	0.00404

Clustered (hour) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

OLS with cyclist FE and day-of-week FE. SEs clustered by hour. Columns 1–2: replication (2013–2017), 3–4: extension (2013–2025), 5–6: post-2017 (2018–2025). Odd columns: all distances; even columns: trips $\geq$2 km.

3.5 Robustness

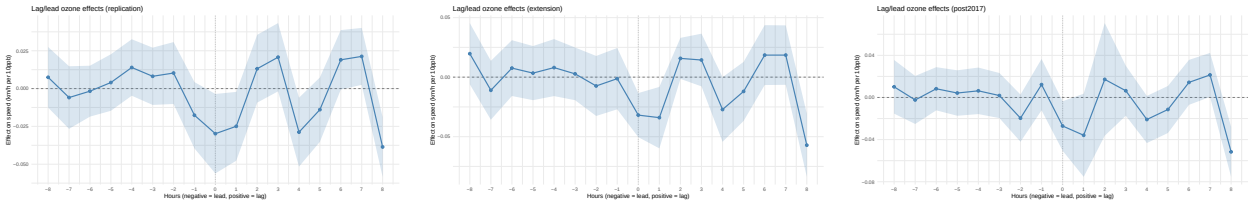
Non-linearity. Figure 1 plots the within-day coefficients on 5 ppb ozone dummies. The replication panel confirms the threshold around 20 ppb documented by Klingen and van Ommeren (2020). The pattern persists in the post-2017 data.



(a) 2013–2017 (b) 2013–2025 (c) 2018–2025

Figure 1: Non-linear ozone effects (5 ppb dummies, within-day specification).

Lag and lead effects. Figure 2 shows the lag/lead analysis. The contemporaneous effect is the strongest, with some persistence in the first lag reflecting ozone autocorrelation. Lead coefficients are near zero, supporting the causal interpretation.



(a) 2013–2017 (b) 2013–2025 (c) 2018–2025

Figure 2: Lag and lead ozone effects (±8 hours, within-day specification).

Median speed. Using median rather than mean speed as the dependent variable yields within-day coefficients of −0.034 (replication), −0.045 (extension), and −0.041 (post-2017), confirming that outliers do not drive the results.

4 Comparison with the Original Study

Table 6 summarizes the key ozone coefficients from Klingen and van Ommeren (2020) alongside our replication and extension estimates. The coefficient represents the effect of a 10 ppb increase in ozone on cycling speed (km/h).

Table 6: Comparison of ozone coefficients with Klingen and van Ommeren (2020).

Strategy	Original (2013–2017)	Replication (2013–2017)	Extension (2013–2025)	Post-2017 (2018–2025)
Within-day	−0.053***	−0.047***	−0.057***	−0.050***
Between-day	−0.083***	−0.246***	−0.111***	−0.076**
Spatio-temporal (week)	−0.044	−0.049	−0.075***	−0.089**
Panel ($d \geq 0$)	−0.026 to −0.031	−0.014***	−0.009***	−0.007**

Original estimates from Klingen and van Ommeren (2020): within-day from their Table 2 col. 1; between-day from Table 2 col. 2; spatio-temporal from Table 3 col. 2 (zone-week); panel range from Table 4. Our replication omits traffic controls; the original study showed these to be inconsequential. Our panel confines a pseudo-cyclist to one year-month and excludes electric bikes. ***, **: significant at 1%, 5%.

Several patterns emerge from the comparison. First, the within-day replication (−0.047) is close to the original (−0.053), confirming that omitting traffic controls does not materially

affect the estimate, as the original authors demonstrated. Second, the within-day effect is stable over time (-0.050 in the post-2017 period), despite higher ozone levels (25.2 vs. 18.7 ppb mean). Third, our between-day replication (-0.246) exceeds the original (-0.083); this difference likely reflects different weather data sources (Open-Meteo vs. Met Office) and the absence of traffic controls, to which the between-day specification may be more sensitive. Fourth, the spatio-temporal estimates, while imprecise in the replication period, become significant in the extended and post-2017 samples as statistical power increases. Finally, the panel estimates (-0.014 to -0.007) are smaller than the original range and decline over time, but remain statistically significant, so ozone also reduces speed within pseudo-cyclists.

5 Conclusion

We replicate and extend Klingen and van Ommeren (2020) using 112 million London bike-share trips spanning 2013–2025. The core finding—that ambient ozone reduces cycling speed at concentrations well below regulatory standards—holds in all four identification strategies, although the spatio-temporal estimate is imprecise for 2013–2017. The within-day estimate is stable across periods. The within-cyclist estimate is about half the original one once pseudo-cyclists are confined to one year-month, and is smaller after 2017.

The persistence of this effect through COVID-19, ULEZ expansion, and cycling infrastructure growth—and across a period in which NO_x fell by half while ozone rose by a third—reinforces that it reflects a fundamental physiological response to ozone exposure rather than a statistical artifact of a particular period or pollution mix.

References

Klingen, Joris and Jos van Ommeren, "Urban air pollution and time losses: Evidence from cyclists in London," *Regional Science and Urban Economics*, 2020, *81*, 103504.